

EXPERIMENT 1:

DIODE CHARACTERISTICS AND APPLICATIONS

Subject: Electronic Engineering III
Hamm-Lippstadt University of Applied Sciences,
Lippstadt

Location: L3.3-E02-080

Scheduled day for the experiment: 4.12.2024 and 5.12.2024

The pre-report must be submitted immediately before conducting the experiment

The final report is due on **January 5, 2024.**

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Experiment Objectives

- Understand the operational characteristics of diodes under different conditions.
- Analyze the current-voltage (I-V) curve of diodes through practical measurements.
- Design and test basic diode circuits, including half-wave clippers and rectifiers.
- Perform theoretical simulations to validate the experiments.
- Enhance proficiency with laboratory instruments, including oscilloscopes and function generators.

I. INTRODUCTION

Diodes are fundamental components in electronic engineering, widely used in rectification, signal modulation, and protection circuits. This experiment aims to explore their characteristics and applications through practical implementations and analysis.

The components and devices used in this experiment, such as diodes, potentiometers, voltage sources, current sources, resistors, multimeters, and breadboards, are detailed in Section II. Section III describes four main tasks:

- 1) **Task 1:** Validation of a basic electronic circuit and analysis of the diode's I-V characteristics.
- 2) **Task 2:** Investigation of the Zener diode's operating characteristics.

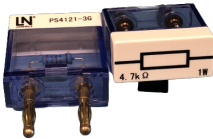
- 3) **Task 3:** Development of a half-wave clipper circuit using a diode.
- 4) **Task 4:** Design and implementation of a half-wave rectifier circuit.

II. COMPONENTS AND DEVICES

• Resistor

Resistors are passive electrical components that resist the flow of current, dissipating energy as heat. The resistance values of the THT resistors used in this experiment are indicated by the color bands on their surface. Table I presents the symbol and a photograph of the resistors.

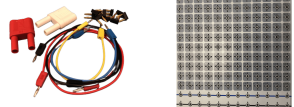
TABLE I: Resistor symbol and device photo

Symbol	Device Photo
	

• Conductor (Wires and Breadboard)

A conductor is a material that allows charge to flow through it. It is used to connect circuit elements in an electrical circuit. The conductors commonly are made of metals, as they have low resistance and high conductance. Breadboard, which is a board widely used as a prototype, acts as a conductor and the construction base of electronics. The breadboard allows the quick creation of circuits without making permanent connections. It is a widely used tool to learn electrical circuits. Table II shows the symbol and the photo of the wires and breadboard. Table II shows the symbol and the photo of the wires and breadboard.

TABLE II: Conductor symbol and device photo

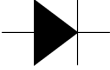
Symbol	Device Photo
	

• Diode

Diodes are semiconductor devices that allow current to flow in one direction. In this experiment, diodes are connected in series to maintain a constant DC voltage across the combination. This configuration ensures that the output voltage remains stable despite load current or supply voltage variations. Table

III shows the symbol and a photograph of the diodes used.

TABLE III: Diode symbol and device photo

Symbol	Device Photo
	

• Oscilloscope

The oscilloscope is an electronic test instrument that graphically displays voltage variations over time. It allows analysis of waveform properties, including amplitude, rise time, and period. Table IV presents the symbol and a photograph of the oscilloscope.

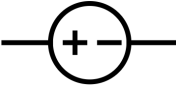
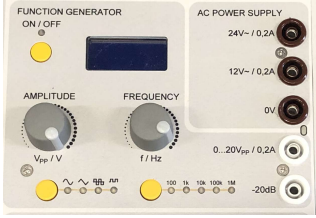
TABLE IV: Oscilloscope symbol and device photo

Symbol	Device Photo
	

• Function Generator

A function generator produces standard waveforms such as sine, square, and triangular waves. The instrument allows precise control over parameters such as amplitude and frequency, making it essential to test and analyze electronic circuits. Table V shows the symbol and a photograph of the function generator.

TABLE V: Function Generator symbol and device photo

Symbol	Device Photo
	

The tasks are structured to reinforce theoretical knowledge with practical implementation and observation, emphasizing both circuit design and signal analysis.

A. Task 1: Diode Characteristics

This task aims to familiarize participants with the behavior of semiconductor diodes by analyzing their volt-ampere (I-V) characteristics. These characteristics reveal the relationship between voltage and current and are critical to understanding how a diode operates in an electronic circuit.

A diode is one of the simplest electronic components, as it allows current to flow in only one direction. Unlike a resistor, which exhibits a linear relationship between voltage and current (Ohm's law), a diode has a nonlinear, exponential I-V curve. As a passive component, a diode does not amplify power. Its operation is divided into two main regions:

- Forward-biased region: The diode conducts current after surpassing a threshold voltage.
- Reverse-biased region: The diode blocks current flow, except for minimal leakage, until the breakdown voltage is reached.

Understanding these regions is essential not only for diodes but also for other semiconductor devices, such as bipolar-junction transistors (BJTs) and field-effect transistors (FETs), where the PN junction serves as a fundamental building block.

1) *Circuit Diagram of Task 1:* The circuit shown in Figure 1 consists of one diode, a voltage source generating a sinusoidal waveform, and a resistor. This setup will be used to observe and analyze the diode's I-V characteristics.

- 1) **Circuit Assembly:** Assemble the circuit as shown in Figure 1. Ensure all components, including the

III. EXPERIMENTS

This section describes the experiments conducted to analyze various aspects of diodes and their applications. Each task is designed to focus on a specific characteristic or use case of diodes, progressing from basic principles to more complex circuit implementations. The experiments include:

- **Task 1:** Analysis of the voltage-current characteristics of a standard diode.
- **Task 2:** Investigation of the Zener diode and its behavior as a voltage regulator.
- **Task 3:** Implementation of a half-wave clipper circuit to limit voltage levels.
- **Task 4:** Construction and analysis of a half-wave rectifier circuit.

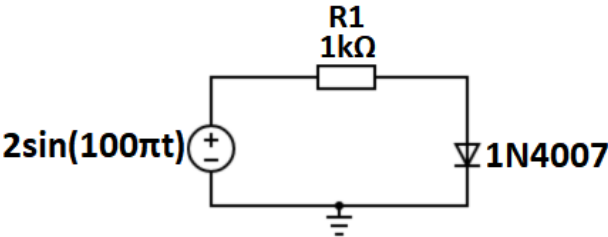


Fig. 1: Circuit of Diode Analysis

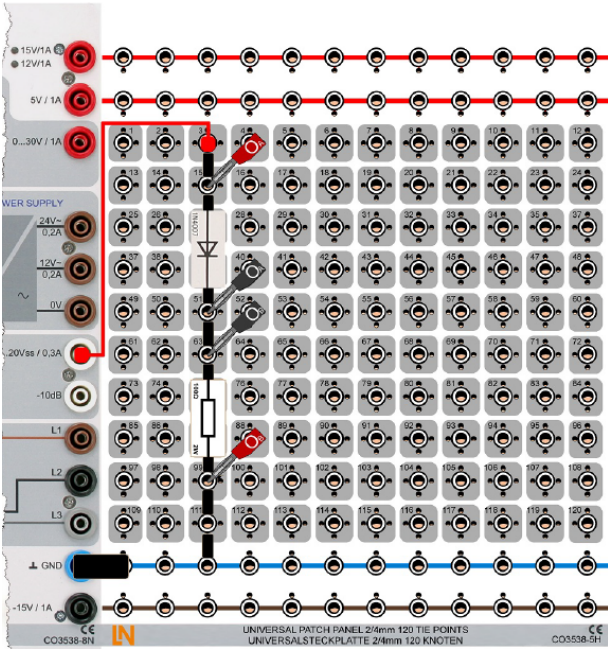


Fig. 2: Circuit Connection with component of Diode Analysis

- 1 kΩ resistor and the 1N4007 diode, are connected correctly as per the diagram. Double-check the diode’s polarity to avoid incorrect behavior.
- 2) **Signal Generator Configuration:** Configure the waveform generator to produce a sinusoidal signal with the following parameters:
- Peak-to-peak voltage (V_{pp}): 4 V.
 - Frequency (f): 50 Hz.
- 3) **Oscilloscope Connections:** Connect the oscilloscope probes to the circuit as follows:
- Channel 1 (X-axis): Voltage across the diode (cathode to anode).
 - Channel 2 (Y-axis): Voltage across the resistor (proportional to the current through the circuit).
- Note:** Since the resistor is in series with the diode, the voltage across the resistor is directly proportional to the current through the diode ($I = V_R/R$).
- 4) **Oscilloscope Settings:** Switch the oscilloscope to

Equipment

The following equipment is needed for this experiment and should be configured as shown:

Equipment	Settings	
	Channel A	Channel B
	Sensitivity	1 V/DIV
	Coupling	DC
	Polarity	norm
	Y position	0
	Time base	4 ms/DIV
	Mode	X/T
	Trigger channel	A
	Trigger edge	pos
	Waveform	sine
	Amplitude	4 V _p
	Frequency factor	1
	Frequency	50 Hz

Fig. 3: Oscilloscope and waveform generator settings for Diode Analysis

- X-Y mode to plot the diode’s I-V curve:
- Navigate to the Menu and select App > X-Y Graph.
 - Set Channel 1 (X-axis) for the diode’s voltage.
 - Set Channel 2 (Y-axis) for the resistor’s voltage (representing current).
 - Adjust the scales of both axes to ensure a clear and complete display of the I-V characteristics.
- 5) **Observation and Analysis:** Use the oscilloscope to analyze the diode’s I-V curve:
- **Forward-biased region:** Observe the exponential increase in current as the voltage rises above the diode’s threshold voltage.
 - **Reverse-biased region:** Verify the minimal current flow and observe if the breakdown voltage is reached.
 - Confirm the expected diode behavior based on theoretical predictions.
- 6) **Data Saving:** Capture and save the I-V curve displayed on the oscilloscope to a USB stick for inclusion in the final report.

B. Task 2: Zener Diode Characteristics

In this task, we analyze the voltage-current (V-I) characteristics of a Zener diode, focusing on its behavior in both forward-biased and reverse-biased regions. Additionally, we perform a load line analysis to determine

the optimal operating voltage and current for the Zener diode when used as a voltage regulator.

1) *Circuit Diagram of Task 2:* The circuit shown in Figure 4 consists of a Zener diode, a resistor, and a sinusoidal voltage source. This setup will be used to observe the diode's V-I curve and analyze its reverse breakdown behavior.

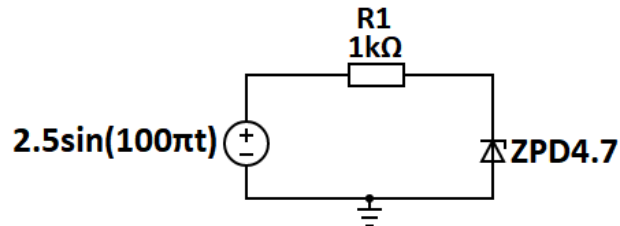


Fig. 4: Zener Diode Analysis Circuit

2) Procedure:

- 1) **Circuit Assembly:** Assemble the circuit as shown in Figure 4. Replace the standard diode from Task 1 with a Zener diode (e.g., 1N4733). Ensure all components, including the $1\text{ k}\Omega$ resistor, are correctly connected as per the diagram. Double-check the polarity of the Zener diode.
- 2) **Signal Generator Configuration:** Configure the waveform generator to produce a sinusoidal signal with the following parameters:
 - Peak-to-peak voltage (V_{pp}): 5 V.
 - Frequency (f): 50 Hz.
- 3) **Oscilloscope Connections:** Connect the oscilloscope probes to the circuit as follows:
 - Channel 1 (X-axis): Voltage across the Zener diode (cathode to anode).
 - Channel 2 (Y-axis): Voltage across the resistor (proportional to the current through the circuit).

Note: The voltage across the resistor (V_R) is proportional to the current through the Zener diode ($I = V_R/R$).
- 4) **Oscilloscope Settings:** Switch the oscilloscope to X-Y mode to plot the Zener diode's V-I curve:
 - Navigate to the Menu and select App > X-Y Graph.
 - Set Channel 1 (X-axis) for the Zener diode's voltage.
 - Set Channel 2 (Y-axis) for the resistor's voltage (representing current).
 - Adjust the axes to clearly observe both forward-biased and reverse-biased regions of the Zener diode.
- 5) **Observation and Analysis:** Use the oscilloscope to analyze the Zener diode's V-I curve:

- Forward-biased region: Observe the exponential increase in current as voltage rises.
- Reverse-biased region: Identify the breakdown voltage, where the diode starts conducting significantly in reverse.
- Confirm the expected Zener diode behavior as a voltage regulator in the reverse breakdown region.

- 6) **Data Saving:** Save the V-I curve displayed on the oscilloscope to a USB stick for inclusion in the final report.

C. Task 3: Half-Wave Clipper Circuit

A half-wave clipper circuit is an electronic circuit designed to clip or limit the voltage of an input signal in one direction only (positive or negative). These circuits are commonly used for signal conditioning, noise reduction, and protection against excessive voltage levels in specific applications.

In this task, we will construct and analyze a half-wave clipper circuit using a Zener diode and a rectifier diode to observe its behavior in limiting the input signal.

1) *Block Diagram of Task 3:* The circuit shown in Figure 5 consists of two diodes: a Zener diode (ZPD4.7) and a rectifier diode (1N4007), along with a resistor and a sinusoidal voltage source. This setup limits the output signal in one direction, depending on the orientation and breakdown voltage of the Zener diode and the forward voltage drop of the rectifier diode.

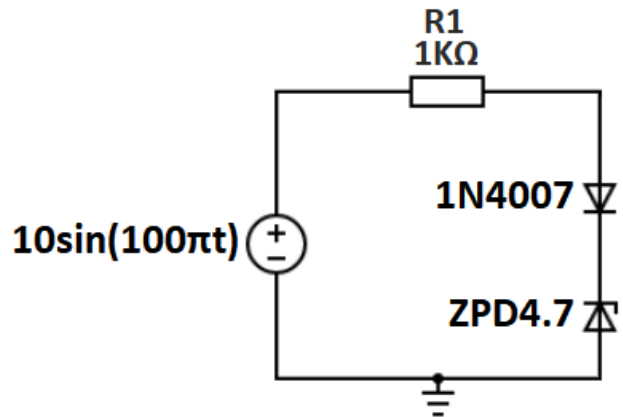


Fig. 5: Half-Wave Clipper Circuit

2) Procedure:

- 1) **Circuit Assembly:** Assemble the half-wave clipper circuit as shown in Figure 5. Ensure all components are connected correctly with proper orientation:
 - Zener diode (ZPD4.7): Positioned to limit voltage during reverse breakdown.
 - Rectifier diode (1N4007): Positioned to conduct during forward bias.

Verify the resistor value (1 k Ω) matches the requirements.

- 2) **Signal Generator Configuration:** Configure the waveform generator to produce a sinusoidal signal with the following parameters:
 - Peak-to-peak voltage (V_{pp}): 20 V.
 - Frequency (f): 50 Hz.
- 3) **Oscilloscope Connections:** Connect the oscilloscope probes to the circuit as follows:
 - Channel 1: Input voltage (across the voltage source).
 - Channel 2: Output voltage (across the clipper circuit).
- 4) **Oscilloscope Settings:** Configure the oscilloscope to display the input and output signals simultaneously:
 - Set Channel 1 for the input waveform and Channel 2 for the clipped output waveform.
 - Adjust the time and voltage scales to ensure the complete waveform is visible.
- 5) **Observation and Analysis:** Analyze the behavior of the half-wave clipper circuit:
 - Observe the clipping effect on the output signal. Verify that the voltage is limited to the breakdown voltage of the Zener diode and the forward voltage of the rectifier diode.
 - Compare the input and output waveforms to confirm the circuit's operation as a half-wave clipper.
 - Note any discrepancies or unexpected behavior and provide explanations based on circuit theory.
- 6) **Data Saving:** Capture and save the oscilloscope output showing both input and clipped signals to a USB stick or other device for inclusion in the final report.

D. Task 4: Half-Wave Rectifier (HWR) Circuit

A half-wave rectifier (HWR) is a simple electronic circuit used to convert an alternating current (AC) voltage into a pulsating direct current (DC) voltage. It allows only one half (either positive or negative) of the input AC waveform to pass through, effectively removing the opposite portion of the signal. This process, called "rectification," is commonly used in power supplies to convert AC to DC.

In this task, you will design, construct, and analyze a half-wave rectifier circuit using a diode and capacitor for filtering. The goal is to observe and explain the behavior of the rectified signal.

1) **Circuit Diagram of Task 4:** The circuit shown in Figure 6 consists of a diode (1N4007), a resistor, a sinusoidal voltage source, and a capacitor for smoothing the output signal.

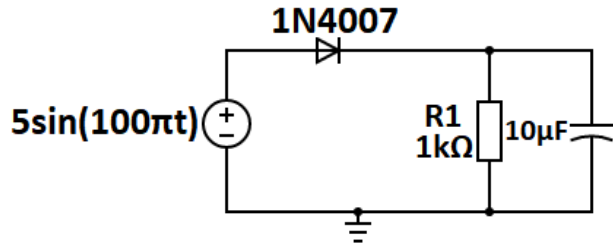


Fig. 6: Half-Wave Rectifier Circuit

2) Procedure:

- 1) **Circuit Assembly:** Assemble the half-wave rectifier circuit as shown in Figure 6. Use the following components:
 - Diode: 1N4007
 - Resistor: 1 k Ω
 - Capacitor: 10 μ F (for output smoothing)
 Ensure the diode is oriented correctly (cathode connected to the resistor) to rectify the positive half of the AC waveform. Verify all connections before proceeding.
- 2) **Signal Generator Configuration:** Configure the waveform generator to produce a sinusoidal signal with the following parameters:
 - Peak-to-peak voltage (V_{pp}): 10 V.
 - Frequency (f): 50 Hz.
- 3) **Oscilloscope Connections:** Connect the oscilloscope probes to measure the input and output signals:
 - Channel 1: Input voltage (across the voltage source).
 - Channel 2: Output voltage (across the resistor and capacitor).
- 4) **Oscilloscope Settings:** Configure the oscilloscope to display the input and output waveforms simultaneously:
 - Set Channel 1 to monitor the input sinusoidal waveform.
 - Set Channel 2 to observe the rectified output signal (before and after smoothing by the capacitor).
 - Adjust the time and voltage scales to clearly visualize both signals.
- 5) **Observation and Analysis:** Analyze the rectifier's performance:
 - Observe the rectified signal on Channel 2 (before filtering) and verify that only the positive half of the input waveform passes through.
 - Observe the filtered signal on Channel 2 (after the capacitor) and confirm that the ripple voltage is reduced, producing a smoother DC signal.
 - Compare the input and output waveforms to understand the rectification and smoothing pro-

cesses.

- Explain any discrepancies between the theoretical and experimental results.
- 6) Data Saving: Save the oscilloscope output showing both the input and output signals (before and after filtering) to a USB stick or other device for inclusion in the final report.

REFERENCES

- A. R. Hambley, *Electrical Engineering Principles and Applications*, 7th Ed, Pearson, 2018.
- A. S. Sedra, K. C. Smith, T. C. Carusone and V. Gaudet, *Microelectronic Circuits*, 8th Ed, Nov. 2019.
- A. Malvino and D. Bates, *Electronic Principles*, 8th Ed, McGraw-Hill, 2016.
- A. University of California Berkeley College of Engineering Department of Electrical Engineering and Computer Sciences, Experiment: Bipolar Junction Transistor Characterization.

Pre-Report for Electronic Engineering III - Experiment 1: Diode

Group:

Date:

Student Identification

Matriculation Number	Student Name
2219941	Md Mostafizur Rahman
2220046	Ikramul Hasan Sazid

Pre-Report Activities

- Perform simulations for Tasks 1, 2, 3, and 4 using any appropriate software.
- Predict theoretical values for the parameters listed in the measurement table and fill in the corresponding column.
- The goal is to compare these theoretical predictions with the experimental results obtained during the lab session.
- Submit the pre-report before the experiment, attaching this page along with the completed simulation results.

Measurement Parameters by Task

Task	Parameter	Description	Measured Value
1	Forward Voltage (V_f)	Voltage across the diode in the forward-biased region.	0,6
1	Forward Current (I_f)	Current through the diode in the forward-biased region.	1,39mA
2	Breakdown Voltage (V_z)	Voltage at which the Zener diode begins conducting in reverse.	4,68V
2	Reverse Current (I_z)	Current through the Zener diode in the reverse-biased region.	0A
3	Clipping Voltage (V_{clip})	Voltage at which the signal is clipped in the positive and negative cycles.	+4,7, -10
4	Output Voltage (Rectified, Peak)	Peak voltage of the rectified signal at the output of the half-wave rectifier.	4,37V
4	Ripple Voltage (V_{ripple})	Voltage variation (ripple) in the rectified signal after filtering.	3,5V

TABLE VI: Measurement Parameters Organized by Task

Grades for Simulations

N_1	N_2	N_3	N_4	N_F

Notes:

- N_1 : Grade for Task 1 simulation.
- N_2 : Grade for Task 2 simulation.
- N_3 : Grade for Task 3 simulation.
- N_4 : Grade for Task 4 simulation.
- N_F : Final grade, calculated as the average of the four tasks:

$$N_F = \frac{N_1 + N_2 + N_3 + N_4}{4}$$